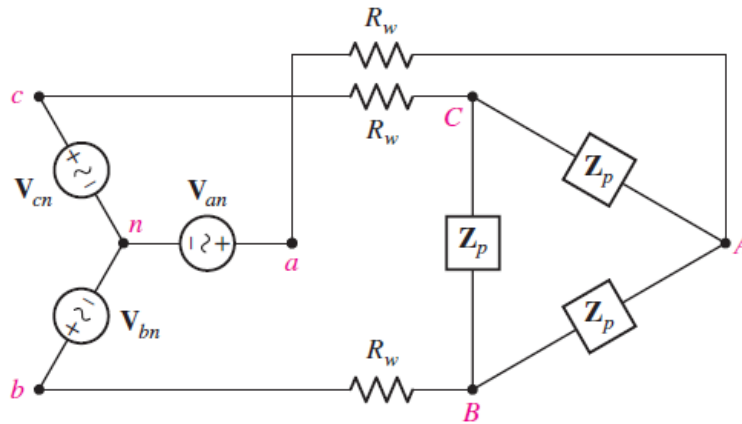


Set A:

The balanced Δ -connected load in the Figure is demanding 10 kVA at a lagging PF of 0.91. If line losses are negligible, calculate I_{bB} and V_{an} if $V_{ca} = 160 \angle 30^\circ$ V and the source voltages are described using the positive phase sequence.



Set B:

Compute I_{aA} , I_{AB} , and V_{an} if the Δ -connected load of the Figure draws a total complex power of $1800 + j700$ W, $R_w = 1.2 \, \Omega$, and the source generates a complex power of $1850 + j700$ W. Assume positive phase sequence.